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**SUPPLEMENTARY MATERIAL TO THE DISSERTATION “MULTITASK
LEARNING FOR POINT-OF-INTEREST CLASSIFICATION AND PREDICTION
TASKS: THE ROLE OF THE CHECK-IN-LEVEL REPRESENTATION”**

**FLORESTAL - MINAS GERAIS
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About this volume

This volume holds five appendices that were written for the dissertation named on the cover of this volume and that are published beside it rather than inside it. The first records every departure of the three reproduced article chapters from their published or submitted originals. The second measures how much of the next category can be predicted from the category history of the input window alone, which is the reference level that one screening argument in the dissertation is read against. The third states the author’s position on whether review by a research ethics committee was required for this work. The fourth records how one hyperparameter of the place-embedding baseline was selected before that baseline was applied to the datasets studied here. The fifth widens the dedicated next-category model until it carries the joint model’s parameter budget, holding the representation, the folds, and the training protocol fixed, and finds the wider model scoring below the narrower one it replaces at the dataset where it ran the full four-seed protocol.

The first two were part of the main document until July of 2026. They were moved here on the advisor’s assessment that a reproduced chapter need not be an identical copy of the article it reproduces, and that the short preface opening each of those chapters already tells the reader what was changed and why. That assessment settles the question the errata appendix was written to answer, at least for the reader of the dissertation. It leaves the underlying record worth keeping, because a departure from a published text should be documented somewhere a reader can reach, and this volume is where it is documented.

The last three left the main document in August of 2026, on the author’s judgment that each answers a question a reader may bring to the work without being needed to follow its argument. They are published here for the same reason the first two are. A record that a reader can reach is worth more than a record that is left out because it did not fit the shape of the chapter it would have interrupted.

The dissertation is complete without this volume. Nothing here is required to follow its argument, to read any of its results, or to judge the work: every claim the dissertation makes is stated and supported in the dissertation itself. What this volume adds is auditability. A reader comparing a reproduced chapter against the article it came from will find the differences enumerated here instead of having to reconstruct them, and a reader who wants the fuller treatment of the label-history benchmark will find it here rather than in the paragraph that summarizes it. A reader who asks how the human-subjects question was handled, or how the baseline was adapted before it was used will find each of those answered here as well.

Two conventions follow from the split. The first two appendices keep the letters they carried when they were part of the dissertation, B and D, so that the records and correspondence

that name them by letter still point at the same material, while the two that arrived later carry E and F, letters of this volume alone; the dissertation letters its own appendices independently, so an appendix letter should always be read together with the volume it belongs to. And a reference in these pages to a numbered chapter, section, table or figure is a reference into the dissertation, not into this volume: this volume numbers only its own five appendices and their tables.

Appendix

APPENDIX B – Errata to the Reproduced Articles

Chapters 3 to 5 reproduce the three articles of this collection in a unified format. Purely structural changes of the re-typeset, such as renumbered sections and labels, tables re-set without vertical rules, captions placed above tables, one-column figure sizes, and the substitution of each article’s textual self-references by chapter self-references, are declared in Section 1.5 and are not listed individually. This appendix lists every *content* departure from the published or submitted texts: corrections of defects known in the article records, wording substitutions required by the dissertation’s writing rules, and the corrections applied in the global bibliography. Each article folder in the research repository keeps the corresponding errata registry; the published records themselves are not edited.

None of the three chapters reproduces the front matter of its article. The abstract and the keyword list are omitted in all three, the Portuguese resumo of the CoUrb article is omitted with them, and the acknowledgments sections of the CBIC and CoUrb articles are omitted as well; the MobiWac manuscript carries none. Venue, pages, identifier, status, and authorship are stated in each chapter’s preface instead, and the acknowledgments belong to the article records rather than to the chapter bodies.

B.1 Article 1 (Chapter 3, CBIC 2025)

Table 1 lists the content corrections applied in the chapter text.

Table 1 – Content errata of the CBIC 2025 article and the corrections applied in Chapter 3. Numeric values quoted in the corrections are the published table values.

Defect in the published article	Correction in the chapter
Prose states “almost four times more wall time”; the convergence table gives 80.88 s for the joint model against a cumulative 34.97 s for the single-task models, a factor of about 2.3.	Prose reconciled to the table: “about 2.3 times the cumulative 34.97 s of the individual single-task models”.

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Table 1 continued from the previous page.

Defect in the published article	Correction in the chapter
Prose states the MFLOPs cost of the joint model is “roughly double”; the table gives 0.234 MFLOPs for the joint model against 2.315 and 0.012 for the two single-task models.	Prose reconciled to the table: the text now states that the table does not show a higher MFLOPs cost for the joint model and quotes the three cell values; four follow-on wording edits restrict the overhead claim to wall time.
Broken cross-reference: the label intended for the single-task heads sits on the Dataset subsection, and the method section points to it as if it described the heads.	Dataset subsection relabeled; the dangling pointer sentence was removed, since no section describing the single-task heads exists in the published paper.
Typo “spatio-tegm mporal”.	“spatio-temporal”.
Unfilled dataset placeholders (N_{users} , N_{poi} , N_{checkins}) in the results section.	Filled with the Florida figures of record for the same Gowalla data, as published in the CoUrb dataset table: 20,301 users, 65,009 POIs, and 990,518 check-ins. The published CBIC paper left these three statistics as placeholders; they are taken from the published CoUrb table so that one corpus figure serves both chapters, and they are the counts that the extraction pipeline of that time produced, not the counts the current pipeline produces (Section B.4).
The optimizer subsection states that Nash-MTL’s “reliance on gradient signs rather than scales inherently balances heterogeneous tasks”. Nash-MTL does not read gradient signs, and the same subsection states the correct property four paragraphs earlier, listing scale invariance among the optimizer’s axioms.	Corrected to name the property the method has: its invariance to the scale of the individual task gradients, the axiom already listed in the same subsection, balances heterogeneous tasks without manual reweighting. Checked against the optimizer’s own paper and against the implementation released with the two studies, neither of which uses gradient signs. The claim is weaker than the published one, and no result depends on it.

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Table 1 continued from the previous page.

Defect in the published article	Correction in the chapter
Prose states that Nash-MTL “requires only two matrix-vector products per iteration”.	The clause is removed from the reproduced sentence and its correction given in a footnote. The claim is not supported by the method’s own paper, where the phrase does not occur, and it understates the cost: both implementations examined run an iterative concave-convex procedure, twenty passes by default, each a convex solve, on top of one backward pass per task. The correction raises the reported cost of a method this chapter used, so it runs against the chapter’s own interest.
The rationale for hard parameter sharing states that the approach “frequently matches or exceeds the performance of more complex architectures on many benchmarks, while offering faster training and inference”, citing a work on task grouping.	Narrowed to the inference-cost claim the cited work supports, that a single network is evaluated rather than one network per task, with the correction recorded in a footnote. The cited work names improved accuracy and reduced training time among the benefits joint training may have in theory, then reports empirically that joint training often performs worse than separate networks, and its contribution is a framework for splitting tasks across networks, which it describes as costly at training time. Its one result favoring joint training is conditioned on a matched parameter budget and on three or more tasks, and it excludes the two-task case this chapter studies. The correction removes a stated advantage of the architecture this chapter adopts, so it runs against the chapter’s own interest.
The survey of optimization techniques states that MGDA “finds Pareto-optimal descent directions”. Pareto optimality and a per-step descent direction are two different guarantees, and the cited work proves the first for a bound rather than for the step.	Narrowed to the two properties the cited work states. Its method computes a direction that decreases every task loss, or else the current point is already Pareto-stationary; the Pareto optimality it proves is for an upper bound on the multi-objective loss and holds under stated assumptions. Neither the chapter’s design nor any of its results depends on this method, which it does not use.

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Table 1 continued from the previous page.

Defect in the published article	Correction in the chapter
The rationale for hard parameter sharing attributes the claim that constraining the hypothesis space regularizes, and generalizes better when tasks are related, to a work on learned soft sharing. That work is a method for learning what and how much to share, presented as an improvement on fixed sharing, so it argues against the sentence rather than for it.	Re-attributed to the work that states the argument the sentence makes: choosing a hypothesis space large enough to contain a solution yet small enough to generalize, searched over an environment of related tasks. The substituted reference was already in the bibliography and is cited for the same claim elsewhere in the chapter. The soft-sharing work remains cited where the chapter discusses cross-connection architectures and negative transfer.

Two elements of the published article were deliberately preserved rather than corrected. The sentence stating that Nash-MTL “consistently yielded a better overall performance” is reproduced verbatim; it predates a later finding about the optimizer implementation, and the chapter preface carries the of-the-time caution instead of an edit. The emphasis convention of the published category table, which bolds the better of the joint and single-task models per row and never bolds the HMRM baseline, is also preserved exactly; the new caption states the convention so the bolding is not misread as “best per row”.

The chapter also carries marked additions that are not part of the published text, each declared as an addition in the chapter source. Two are frame devices required by the dissertation: the italic preface, which states the venue, the identifier, and the of-the-time framing, and the lead sentence that introduces the convergence table, written because the dissertation’s presentation rules require a table to be introduced by a takeaway sentence. The others state protocol facts the published paper left implicit: that the folds are stratified by sample rather than by user, so the check-ins of one user may appear in both training and validation; that the sample unit of the category task is the place, so no place spans two folds; that the code of record pins a single random seed, so the five folds are one repetition of the experiment and the reported standard deviations are the spread across those folds at that seed; and that training runs for the full configured number of epochs without early stopping, with each task read at the epoch of its own highest validation macro-F1 on the fold the score is reported on. All four are recovered from the code released with the article and none is a claim about a result. The number of model configurations examined during the study is not recoverable from the released material, and no sentence of the chapter asserts one.

Table 2 lists the wording substitutions applied to the preserved prose under the dissertation’s writing rules. Most are grammatical or terminological corrections that change neither the strength nor the scope of a claim. Four remove the words *significant*, *significantly*, and *statistically* from statements about the chapter’s own measured comparisons: the chapter reports

no inferential test of any kind, so these words claimed more than the study established, and the replacements state the same findings descriptively. Four further uses of the word were left in place. Two report claims of cited work, one states the study’s own hypothesis, and one names a chosen target level. Two more remain that do read against this chapter’s own results: one describes the multitask model as incurring a significantly higher overhead in wall time, and one refers to the lack of significant improvement from that model. Both are left as published, because both understate rather than overstate what the chapter shows: they qualify results the chapter reports as negative, so removing the word would make the chapter’s own conclusion sound stronger than it was. A reader should nevertheless read them descriptively, since no inferential test supports either. Another row records a sentence rewritten for readability: the published clause about an “unbalanced result” between the two models had no grammatical referent for what it said this could make worse, and the chapter now states the same possibility in the terms the surrounding paragraph establishes, with its hedge preserved. A further row records a footnote added to a definition rather than a substitution. The table also corrects the reversed names of the two tasks and removes an untested robustness claim about data-sparse environments. No edit in this table raises the strength of a claim.

Table 2 – Wording substitutions in the reproduced CBIC 2025 prose (writing-rule conformance; claim strength unchanged or reduced, never raised).

Published wording	Chapter wording
“By leveraging shared information, MTL can potentially mitigate...”	“By using shared information, ...”
“solutions that leverage shared representations across related POI tasks”	“solutions that use shared representations...”
“by leveraging shared information across correlated tasks”	“by sharing information across correlated tasks”
“Moreover, we utilize a Deep Graph Infomax...” / “Moreover, $\mathcal{D}$ is a discriminator...”	“In addition, ...” (both instances)
“These findings underscore that...”	“These findings indicate that...”
“Furthermore, investigating advanced multi-task optimizers...”	“In addition, investigating...”
“Furthermore, the computational cost ... was roughly double”	Subsumed by the MFLOPs correction in Table 1.
“where different sequential patterns might be leveraged”	“might be exploited”
“MTL allows the learning of complementary objectives simultaneously, allowing for more accuracy”	“MTL allows complementary objectives to be learned simultaneously, which can improve accuracy.”

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Table 2 continued from the previous page.

Published wording	Chapter wording
“Some Models such as TME ... but treat prediction and classification separately”	“Some models, such as TME, ... but treat prediction and classification separately.”
“jointly performs POI category prediction and next-POI classification”	“jointly performs POI category classification and next-category prediction” (the published sentence had the task operations reversed).
“enabling more robust predictions even in data-sparse environments”	Removed because the study did not evaluate robustness as a function of data sparsity.
“We hypothesize three primary factors contributed to this outcome”	“We hypothesize that three primary factors contributed to this outcome.”
“the MTL framework does not deliver the consistent, significant gains often expected”	“does not deliver the consistent gains often expected” (the chapter reports no inferential test, so the word carries no verdict here).
“both our MTL and Single models significantly outperform HMRM across all POI categories”	“outperform HMRM in every POI category” (the comparison is stated by scope instead of by significance).
“significantly surpassing MHA+PE in these categories”	“surpassing MHA+PE in these two categories”.
“suggesting that, statistically, the performance of the MTL and Single models is largely comparable, without a clear, consistent, and significant advantage”	“suggesting that the performance of the MTL and Single models is largely comparable, without a clear or consistent advantage”.
“Also, it is important to notice that since we have an unbalanced result for the MTL and single, this could lead to the worse of other results.”	Rewritten for readability, with the claim unchanged: the chapter states that the two models do not lead consistently across the categories, each obtaining the better F1-score in some of them, and that this uneven distribution could make either model appear worse in the categories where the other one leads. The published phrase “unbalanced result for the MTL and single” names the pattern the two preceding sentences describe, and “the worse of other results” had no grammatical referent. The hedge “could” is preserved, no number is touched, and no inferential claim is added.
“Next-POI Prediction, in contrast, aims to predict which specific location a user is likely to visit next” (the related-work definition of the research area)	Reproduced verbatim, with a footnote added stating that this is the general formulation of the literature and that the variant studied in the chapter predicts the category of the next visited place.

B.2 Article 2 (Chapter 4, CoUrb 2026)

The first two corrections in Table 3 replace the published numbers by the values of an internal audit of the same results, performed after publication; the audited values are the ones the chapter reports. The third substitutes a verdict verb without changing the strength of the claim, and the fourth narrows one claim about a cited method to the guarantee that method’s own paper states. The last three correct attribution and vocabulary: two claims are returned to works that state them, and one data column is named for the reader rather than by its identifier in the source files.

One element was preserved rather than corrected: the published bolding of the Florida *Outdoors* baseline cell in the sequential-task table is kept exactly as published; the technical-tie reading lives in the prose only.

One sentence of the methodology was re-typeset without a change of content. The published sentence introducing the category embedding of the POI Encoder placed a long inline expression at the end of its first line, where the one-column measure of this format cannot break it and the line runs past the margin. The clause order is reversed so that the expression falls earlier in the sentence. Every content word of the published sentence is kept and no quantity, symbol, or claim changes, so this is recorded among the mechanical changes of the re-typeset rather than as a content departure.

The chapter carries marked additions that are not part of the published text, each declared as an addition in the chapter source. Three are frame devices required by the dissertation: the italic preface, a recap subsection on the MTLnet framework of Chapter 3, and the reading instruction appended to the caption of Figure 3, which names the axes and the pattern the panels show so that the figure can be read on its own. The chapter states no protocol fact beyond those the published paper itself reports. The number of model configurations examined during the study is not recoverable from the released material, and no sentence of this chapter asserts one. The evaluation protocol of this study, which stratifies the cross-validation split by sample rather than by user, is stated in the chapter preface, where it frames the study alongside Chapter 3. The remaining three are the lead sentences that introduce the three tables, one each for Tables 5, 6, and 7, written because the dissertation’s presentation rules require a table to be introduced by a takeaway sentence. Each of the three reads its table and states no quantity that the table does not already show.

B.3 Article 3 (Chapter 5, MobiWac 2026)

One content correction was applied. The submitted manuscript describes the prior work of Chapter 3 ([1]) as having studied the next-category and next-region tasks and as having observed negative transfer between them. Both statements are inaccurate against that work:

Chapter 3 pairs static category classification with next-category prediction and contains no region task, and it hypothesizes negative transfer to explain a parity null rather than observing it. The two sentences (in the introduction and the related-work recap) were corrected to state that the earlier work found no consistent multitask advantage for the paired category tasks, and that the present study is the one that introduces the region task. The correction removes an internal contradiction with the chapter’s own novelty claim and does not alter any experimental result; it was applied both here and in the version of record, and is recorded in that article’s errata. Beyond it, every number and claim in Chapter 5 reproduces the submitted text. The chapter also carries three marked additions declared in the chapter source: the italic preface, a recap subsection covering the artifacts of Chapters 3 and 4, and the restoration of one figure that the submitted eight-page build had cut for space, whose four values are stated in the reproduced prose in both versions. This chapter writes no new lead sentences for its tables, since the submitted text already introduces each of them.

Table 4 lists two further departures in the reproduced prose. Neither alters an experimental result. The first narrows a universal claim to the channels its evidence covers; the second adds a cross-reference to the appendix that measures the same quantity on the final model.

This article differs from the other two in a way that changes what this section has to record. It is under review rather than published, so a correction to a minor factual or scope claim does not have to be declared here at all: it is applied to the submitted source as well, and the two texts stay identical. Three corrections were handled that way and are named in the article’s own errata record rather than in this table. What remains here are the two that could not be folded into a paper under review, because each depends on material the article has no room for: one cites this chapter’s own results table, and one cites the gradient-cosine appendix of the dissertation.

B.4 Two extractions of the same Gowalla corpus

Chapters 3 and 4 report the Florida subset of Gowalla as 20,301 users, 65,009 POIs, and 990,518 check-ins, while Chapter 5 reports 21,052 users, 76,544 POIs, and 1,407,034 check-ins for the same state of the same public dataset. The two are different extractions of that state, not different datasets and not a discrepancy between them: the first two chapters report the corpus as the extraction pipeline of their time produced it, and Chapter 5 reports the current extraction.

The mechanism is the category mapping, not the raw data and not a different filtering rule. The extraction assigns a category to each check-in from a mapping table and discards the rows whose category does not map, so widening that table keeps rows that an earlier run had dropped. The mapping files were extended about eleven months after the earlier Florida extraction, and the places they added fall mostly in *Entertainment*, *Outdoors*, and *Travel*. A controlled comparison against the working extraction of the earlier period confirms the direction: each of its POIs, users, and check-ins reappears in the current extraction, which adds further ones. The current

pipeline therefore cannot reproduce the earlier counts, and the two sets of figures are kept as each chapter reported them rather than retrofitted to one another. The comparison is recorded in the research repository, in `src_utils/cbic_recompute_result.md`.

B.5 Bibliography errata

The dissertation uses a single global bibliography. During its consolidation, the entries inherited from the three articles’ reference lists were re-verified against the sources of record (Crossref, OpenAlex, arXiv, publisher pages), and the defects in Table 5 were corrected. In addition, duplicate keys naming the same work across the three donor lists (among them three spellings for the Deep Graph Infomax paper and two for the Nash-MTL paper and for the Gowalla reference) were consolidated to one entry each; key consolidation does not alter any reference’s content and is not itemized.

Table 5 – Bibliography errata: defects in entries inherited from the articles’ reference lists and the corrections applied in the global bibliography. Keys in typewriter font.

Defect in a donor reference list	Correction in the global bibliography
POI-RGNN cited the wrong paper for the Capanema et al. model.	Replaced by the correct record (capanema2023poirgnn: Ad Hoc Networks, v. 138, 103016, 2023), verified against Crossref.
HMRM entry (chen2020modeling): garbled and incomplete author list, journal article typed as conference paper, DOI missing.	Five authors restored per the Crossref record; type corrected to journal article (IEEE TKDE, 34(4):1902–1914, 2022), DOI 10.1109/TKDE.2020.3001025.
Graph Attention Networks cited as an arXiv preprint.	Cited as the ICLR 2018 version of record; arXiv identifier kept as a note.
The spherical-harmonics location encoder of Rußwurm et al. was typed as an arXiv preprint, although the entry already recorded the year of the conference version.	Cited as the ICLR 2024 version of record, confirmed against the arXiv journal reference and the OpenReview submission record; arXiv identifier kept as a note.
Cross-Stitch entry (misra2016cross): DOI resolves to a different paper.	DOI corrected to 10.1109/CVPR.2016.433 (verified).
MTL-survey entry (zhang2021survey): DOI does not resolve.	DOI corrected to 10.1109/TKDE.2021.3070203 (verified).

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Table 5 continued from the previous page.

Defect in a donor reference list	Correction in the global bibliography
The node feature matrix is described as the category one-hot encoding of each POI. The released implementation feeds the mean of the one-hot vectors of a POI’s graph neighbors, with the POI’s own vector excluded.	Published sentence preserved; a footnote records what the code does and why the distinction matters. The embedding therefore describes a neighborhood rather than a POI’s own label, so the static task it supports is spatial homophily.
Kohavi (1995), the cross-validation and bootstrap study: the entry carried a DOI that resolves to a re-deposit of the paper made in 2026, not to the conference record.	DOI dropped, since the IJCAI-95 proceedings predate DOI registration. Author, title, volume and pages 1137–1143 confirmed against the proceedings PDF published by IJCAI.
church2017word2vec resolves to a commentary column, not the skip-gram method.	Replaced by mikolov2013word2vec (arXiv:1301.3781) at the citing sentence.
The sentence corrected in the row above attributes both skip-gram and negative sampling to mikolov2013word2vec (arXiv:1301.3781), which introduces skip-gram but not negative sampling.	mikolov2013negsampling (arXiv:1310.4546) added alongside it, which is where negative sampling is introduced. Both keys now appear at the citing sentence, since it claims both methods.
yu2019mmoe: no distinct resolvable work; its claimed arXiv identifier resolves to an unrelated toolkit paper.	Dropped; the citing sentence describes MMoE and now cites ma2018mmoe alone.
Deep Graph Infomax cited under three different key spellings across the donor lists.	Consolidated to a single entry (velickovic2019deep, ICLR 2019).
silva2025mtlnet (the Chapter 3 article, cited by the CoUrb paper): venue name conflated with another conference, stale “Submetido” note, wrong author list.	Venue corrected to CBIC 2025 (Congresso Brasileiro de Inteligência Computacional), DOI 10.21528/CBIC2025-1191324; note dropped; six authors restored per the Crossref record.
paiva2026stmtlnet (the Chapter 4 article): third author missing in the seed entry.	Author restored per the Crossref record of DOI 10.5753/courb.2026.22960; pages 323–336 confirmed.
standley2020tasks: venue given as CVPR; page range unverifiable.	Venue corrected to ICML 2020 per OpenAlex; arXiv:1905.07553 noted; the unverifiable page range dropped.
kokkinos2016ubernet: typed as an arXiv preprint, with the preprint identifier standing in for the journal field.	Cited as the CVPR 2017 version of record, DOI 10.1109/CVPR.2017.579, pages 5454–5463, verified against Crossref; arXiv identifier kept as a note.

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Table 5 continued from the previous page.

Defect in a donor reference list	Correction in the global bibliography
The Sphere2Vec location encoder of Mai et al. was typed as an arXiv preprint, with no venue recorded at all.	Cited as the journal version of record (ISPRS Journal of Photogrammetry and Remote Sensing, v. 202, pages 439–462, 2023), DOI 10.1016/j.isprsjprs.2023.06.016, verified against Crossref and against the journal reference recorded on the preprint; arXiv identifier kept as a note.
CoUrb chapter (4): the Gowalla dataset was cited via Liu et al. (2014), “Exploiting Geographical Neighborhood Characteristics for Location Recommendation” (ACM CIKM), a location-recommendation method paper, not the dataset source.	Citation corrected to the dataset sources (cho2011gowalla, jure2014snap), which the same chapter already cites for the dataset elsewhere; the location-recommendation reference was dropped from the chapter.
nash (Nash-MTL) recorded the venue by the abbreviation this bibliography uses for that conference, and carried neither the proceedings volume, the series, the publisher, nor a resolvable link. The three articles’ own lists went further and cited the work as a preprint, with a preprint identifier in place of a journal or proceedings field.	Brought to the publisher’s own record for the paper: the proceedings of the thirty-ninth International Conference on Machine Learning, volume 162 of Proceedings of Machine Learning Research, pages 16428–16446, 2022, with the publisher and the publisher’s landing page. The page range was already correct here and is unchanged. The same entry was corrected in each of the three article source trees, where the entry type itself was wrong.

Table 3 – Content errata of the CoUrb 2026 article and the corrections applied in Chapter 4 (English translation of the published text).

Defect in the published article	Correction in the chapter
Results and conclusion state wins over the baseline in “16 of the 21 evaluated combinations”.	Audited count: 15 of the 21 combinations, with one additional technical tie (<i>Outdoors</i> in Florida, where the baseline mean exceeds the best variant by 0.02 percentage points, within one standard deviation).
Introduction, results, and conclusion state “average gains of 20 to 24 percentage points” on the category task.	Audited per-state means: gains of 20.2 to 22.0 percentage points, considering the better of the two spatial encoders in each combination (the disclosure the audit requires).
Introduction states “wins in most scenarios”.	“outperforms the baseline in most scenarios” (verdict-verb rule; claim strength unchanged).
The methodology states that maximizing the product of the task utilities “ensures that the update is beneficial for all tasks simultaneously”. The guarantee is unconditional as written.	Narrowed to the guarantee the method’s own paper derives: at points that are not Pareto stationary, and under its assumption that the task gradients there are linearly independent, the update direction is a descent direction for every task. The paper states the property in those terms, and it approximates the bargaining solution by an iterative procedure rather than solving it exactly. The claim is weaker than the published one, and no result depends on it.
The methodology attributes to a next-POI recommendation method the claim that cyclical mobility regularities carry discriminative information about the functional nature of the visited places. That method models long- and short-term user preference and states nothing about temporal signal revealing what kind of place a visit is.	Split into the part the literature supports and the part it does not. The regularity itself is attributed to the paper that published the check-in collection this chapter uses, already cited elsewhere in the chapter, and the recommendation method is kept for the property it does establish, that models of the next visit read temporal structure explicitly. The claim about place function is withdrawn rather than re-attributed, since no work in the bibliography supports it. The paragraph’s argument, that the original architecture lacks an explicit temporal representation, is unchanged.
The category-encoder loss attributes its hierarchical regularization term, an L2 penalty that pulls a subcategory embedding toward its parent over a known label tree, to a paper on nonlinear dimensionality reduction. The two share a graph Laplacian and nothing else.	Re-attributed to a multitask embedding method for place-category annotation, which regularizes relatedness among categories using a predefined category hierarchy: the same construction the loss implements. The reference was already in the bibliography and already cited in this chapter. No term, coefficient, or reported value changes.
Two sentences of the methodology name a data column by its identifier in the source files, <code>fclass</code> , which is not a term a reader of the chapter can resolve.	Replaced by the registered reader-facing name, <code>fine class</code> , in both sentences, set in roman because it is ordinary English rather than a code token. The parenthetical gloss that introduced the column is kept, so the sentence still tells the reader what a secondary category is.

Table 4 – Claim-scope corrections in the reproduced MobiWac 2026 prose (claim-scope narrowing and comparison-basis disclosure; no experimental result altered).

Submitted claim	Correction in the chapter
The representation “passes no usable information about the test visits, on three grounds”.	The universal negative is replaced by a bounded statement over named channels. The first ground is scoped to the training objective, with the note that each visit’s category is a node input feature; a fourth ground is added, reporting the development audit that probes the forward-edge channel between consecutive visits, with its three limits stated: a linear probe, Florida at one random initialization, and earlier builds of the representation. Every number already in the paragraph is unchanged.
The gradient-cosine paragraph reports the development measurement, four seeds on four Gowalla states, and points only to the mechanism test in the results section.	One sentence is added in the chapter, pointing to the appendix that measures the same quantity on the final model at four of the six datasets and reaches the same conclusion. The sentence cites a dissertation appendix, so it is not carried into the submitted paper: the referent does not exist there. No number, scope, or conclusion in the paragraph changes.

APPENDIX D – A Label-History Benchmark for the Next-Category Task

This appendix answers one question. How much of the next category can be predicted from the category history of the input window alone, without reading any representation at all? The answer fixes how strong a screening argument in Chapter 5 can honestly be.

The screening procedure works as follows. For each candidate representation, a linear model is trained to predict the next category from the vector of the last visit in the window alone, and the macro-F1 it reaches is compared against the score a reference encoder reaches on the same task. A candidate scoring far above that reference encoder is suspected of carrying the label it is supposed to predict, and is discarded. The probe is run twice, once on standardized features and once on raw ones, and either run can disqualify a candidate.

All scores in this appendix are macro-F1 on a zero-to-one scale, and differences between them are quoted in points, where one point is 0.01 on that scale.

Two quantities are involved in that comparison. Both have served as the screen’s reference level at different times, and both have been called the ceiling, which is the confusion this appendix exists to remove. Each is given one name here and keeps it.

The label-history benchmark. The best macro-F1 obtained from the recorded category history of the input window alone. No representation is read and no encoder is involved, so this is a property of the sequence of category labels. It is what this appendix measures.

The clean reference encoder. The encoder that a screened candidate is actually compared against, and the score the screening margin is applied to. Two encoders play this role at Florida, both from the lineage the representation of Chapter 5 descends from: one at 0.4090 macro-F1 on standardized features and 0.4074 on raw ones, and a residual variant at 0.4197 and 0.4182. Where the two are discussed together below, they are named in the plural for that reason.

The benchmark is measured under four predictors that see category labels and nothing else. The first is persistence, which repeats the last visited category and fits no parameters. The second is a multinomial logistic model on a one-hot encoding of that same last category. The third is the same model on the counts of each category over the nine visits of the window. The fourth is the same model on positional one-hot encodings of all nine slots, and it is the only one of the four that can use the order in which the categories occurred. The benchmark is the best score of the four. All of them run under the protocol the screen itself uses: five-fold

cross-validation grouped by user, so that no user’s windows fall on both sides of a split [2, 3], with macro-F1 averaged over the five folds. Macro-F1 counts the seven categories equally [4], which is also why the three logistic models are fitted with balanced class weights: an unweighted fit optimizes a different objective and would understate the benchmark.

Across the five datasets where the inputs were available the benchmark ranges from 0.2800 to 0.3617, well above the majority-class floor in every case. Table 6 gives the four predictors, the benchmark, and the floor for each dataset.

Table 6 – The label-history benchmark for next-category prediction, by dataset. Each predictor column sees only the recorded category history of the nine-visit input window; no representation is read. The bold cell in a row marks the predictor that attains the benchmark, and the benchmark column carries its standard deviation over the five folds. The floor predicts the training portion of each fold’s majority class. All values are macro-F1 on a zero-to-one scale over seven categories, averaged over five user-grouped folds. The row counts are those of the harvested per-dataset input tables this audit reads, which are not the stride-one sliding-window counts reported for the experiments in Chapter 5; the benchmark is a property of the label sequence and is unaffected by that difference, but the two counts should not be compared.

Dataset	Rows	Label-history predictors				Benchmark	Floor
		Persistence	One-hot	Counts	Positional		
Alabama	12,709	0.2800	0.2800	0.2783	0.2791	0.2800 ± 0.0138	0.0727
Arizona	26,396	0.3080	0.3080	0.3191	0.3232	0.3232 ± 0.0163	0.0725
Florida	159,175	0.3417	0.3417	0.3503	0.3617	0.3617 ± 0.0069	0.0566
California	358,302	0.3160	0.3160	0.3148	0.3242	0.3242 ± 0.0057	0.0704
Istanbul	58,075	0.2588	0.2588	0.2947	0.3016	0.3016 ± 0.0058	0.0715

Two readings follow from the table, and they point in different directions.

The screening comparison is unaffected. Its verdicts are relative, since a candidate is discarded when it clears the clean reference encoder by more than the screening margin of three points, and the benchmark enters that arithmetic nowhere. The attention-based encoder that was discarded at Florida clears the clean reference encoder by 8.9 points, so its verdict does not depend on where the benchmark lies.

The absolute reading is the weaker one. At Florida the two clean reference encoders score above the benchmark, 0.4090 and 0.4197 against 0.3617, which Chapter 5 reports as a gap of four to six points, and so does every candidate screened beside them on the raw run of the probe. One candidate falls below the benchmark on the standardized run, a relation-typed graph encoder at 0.3328 that rises to 0.4142 on the raw one. It is not an example of a clean encoder sitting below the benchmark, because it is the candidate that passed this screen and then leaked under a downstream sequence model, which Chapter 5 reports as the reason the linear form is a screen rather than a proof. What the screen bounds is therefore the candidates against each other, not any of them against an absolute standard. That is the weaker of the two claims, and it is the one Chapter 5 makes.

The gap is not by itself evidence of a leak. A per-visit vector legitimately carries more than the previous category. It also carries the place, the neighborhood of that place in the graph, and the hour of the visit, and any of those can predict the next category with no information flowing backward in time.

The benchmark is also not an upper bound. It is the best score of four named predictors, so a different model reading the same category history could score above it. Calling it a ceiling, as the internal screening record does, would assert more than the measurement supports.

Two coverage limits apply. Texas is absent because the check-in graph the computation reads was not retained for that dataset under the shipped representation. Only its embedding outputs remain, and measuring the benchmark there would require re-running the preprocessing. For Istanbul, 196 of the 29,816 places carry more than one category across the record. The reason is not recorded in the data, so none is claimed here.

The table maps each place to its most frequent category. A strict variant that instead drops every window ending at one of those places keeps 55,946 of the 58,075 windows and gives 0.3009, against the 0.3016 of the table.

The computation is released with the code as `<scripts/embedding_eval/autocorrelation_ceiling.py>`. Its outputs are `<autocorrelation_ceiling.json>` and two accompanying comma-separated files, all three under `<docs/results/embedding_eval/re screen_cat/>`. Those file names keep the older word, since they predate the naming settled here.

APPENDIX E – The Human-Subjects Question

The studies are a secondary analysis of two collections that were already public before the work began, and no participant was recruited, contacted, or observed by the author. On that basis the author's position is that review by a research ethics committee was not required. This appendix records that position and its basis. It records no approval and no exemption, because none was sought and none is claimed.

Should the program require a formal determination for secondary analysis of public data, it belongs on file before deposit, and this appendix should then name it.

APPENDIX F – Adaptation of the HGI Baseline

This appendix records one methodological choice that is necessary to interpret the HGI baseline used in the later studies. Huang et al. set the cross-region edge weight to 0.4 in the HGI experiments on Xiamen and Shenzhen [5]. The present work did not transfer that value without evaluation. It selected the weight on Alabama before applying HGI to the United States state graphs.

The sweep evaluated weights 0.4, 0.5, 0.6, and 0.7 with five folds and a budget of 50 epochs. The recorded category F1, on a zero-to-one scale and reported as the mean plus or minus the standard deviation across folds, increased in the same order:

Cross-region weight	Category F1
0.4	0.7388 ± 0.0205
0.5	0.7678 ± 0.0211
0.6	0.7944 ± 0.0186
0.7	0.8186 ± 0.0123

The later studies therefore use 0.7. This sweep supports adaptation of one baseline hyperparameter; it is not evidence about the relative performance of HGI and Check2HGI, and it does not change HGI’s role in the representation lineage.

APPENDIX G – A Parameter-Count Control for Next-Category Prediction

Chapter 5 attributes the next-category results to the check-in-level representation rather than to the size of the model that reads it. This experiment tests the competing explanation. The dedicated category model was made wider until it had approximately the same number of trainable parameters as the complete joint model, holding the representation, the folds, and the training protocol fixed. If capacity were what the category task responds to, the wider model should improve on the narrower one. It does not.

The control covers Alabama under the four-seed protocol and California at one seed, the second limited by a per-fold cost thirty-six times higher there. It is an additional dissertation analysis and does not appear in the Chapter 5 manuscript. This control is reported in the supplementary volume because the main volume’s category results are bounded within half a point, the same band this control’s own effects fall near, so it does not carry an argument there.

G.1 Result at a glance

Table 7 places the three results side by side. At Alabama, where the control ran the full four-seed protocol, the wider dedicated model does not improve on the narrower one it replaces: it scores 0.53 macro-F1 points lower with 6.5 times the parameters, and a paired test over the four seeds separates that difference from zero ($p = 0.0011$). The direction is unanimous at the fold level, all twenty folds included, which is a stronger statement than the seed-level mean alone. The joint model, which carries a comparable budget to the wider one, sits between them and above the wider model by 0.35 points ($p = 0.041$). Capacity is therefore not the variable next-category prediction responds to at this dataset, which leaves the representation as the explanation Chapter 5 gives.

At California the same three arms were measured at one seed, where the three lie within 0.06 points of each other. That is a single-seed reading rather than a result under the four-seed convention, and it is reported as such: no test is applied to it and no average is taken across the two datasets.

Table 7 – Next-category macro-F1 for the original dedicated, wider dedicated, and joint models. The last two columns report wider minus original and joint minus wider, in macro-F1 points. Alabama: four seeds $\times$ five folds per arm; $\pm$ is the standard deviation across seeds. [†]California: seed 0 only, five folds, for all three arms; a single-seed reading, not a result under the four-seed convention.

Dataset	Original	Wider	Joint	Wider – original	Joint – wider
Alabama	30.77 $\pm$ 0.07	30.24 $\pm$ 0.13	30.59 $\pm$ 0.07	−0.53	+0.35
California [†]	35.62	35.56	35.61	−0.06	+0.05

The joint column uses the jointly selected checkpoint, the same convention as Chapter 5’s results table, so the joint value here and the chapter’s cell for the same dataset are the same number. Only the dedicated arms differ between the two, because this control replaces the narrow dedicated model with a wider one.

G.2 Experimental design

The control changes one factor: the hidden dimension of the dedicated next-category model. At Alabama each arm uses four seeds, {0, 1, 7, 100}, and five user-disjoint folds per seed, for 20 fitted models per arm, which is the convention the rest of this document uses. At California one fold of the wider model takes 3,682 seconds against 102 seconds at Alabama, so the four-seed protocol would take about twenty hours there; the wider arm was measured at seed 0 and stopped. The reported value is macro-F1 under the same epoch-selection rule as the arms it is compared against.

Table 8 shows the capacity match. The original model uses a hidden dimension of 256. The wider models reach 100.2% and 101.9% of the joint-model parameter counts, so they do not receive a smaller parameter budget. Relative to the original dedicated model, the wider models carry 6.5 and 8.1 times the trainable parameters.

Table 8 – Trainable parameter counts in the parameter-count control.

Dataset	Joint model	Original dedicated	Wider dedicated
Alabama	4,197,621	644,359	4,207,399 ($h = 672$)
California	5,151,189	644,359	5,249,719 ($h = 752$)

Because width can change the best learning rate, the wider model was given its own small search before the reported runs, and it selects 0.0025 rather than the 0.005 the original model uses at both datasets. Table 9 reports the selected setting per dataset. This search is smaller than the one the original dedicated model received, which is a reason the wider arm is not read as a tuned ceiling.

Table 9 – The selected training setting for the wider dedicated category model at each dataset, with the macro-F1 it reaches. Alabama: four seeds $\times$ five folds, $\pm$ across seeds.
[†]California: seed 0, five folds.

Dataset	Hidden dimension	Batch size	Learning rate	Macro-F1
Alabama	672	2,048	0.0025	30.24 $\pm$ 0.13
California [†]	752	8,192	0.0025	35.56

G.3 Interpretation and limits

At Alabama, parameter count does not account for next-category performance: multiplying the dedicated model’s trainable parameters by 6.5 lowers its macro-F1. The competing explanation this control was built to test is therefore not supported, which leaves the input representation as the factor Chapter 5 identifies. What the control does not do is decompose the joint model: it holds the representation fixed and varies width, so it says nothing about which of the joint model’s own mechanisms contribute to its results.

The scope is narrow and worth stating plainly: next-category prediction only, one wider model per dataset, width rather than depth, a smaller hyperparameter search than the original model received, the full four-seed protocol at Alabama and a single seed at California. The result rules out a simple parameter-count explanation at Alabama and is consistent with the same reading at California, but it is neither a complete capacity study nor a component ablation.

Result files. The per-fold scores of every arm, the driver log with per-seed wall-clock times, and the parsed digests are tracked under <docs/results/closing_data/apxi_v18/>.

References

- [1] SILVA, V. H. O. et al. An investigation into multi-task learning for point-of-interest category classification and next-POI prediction. In: **Anais do XVII Congresso Brasileiro de Inteligência Computacional (CBIC 2025)**. [S.l.: s.n.], 2025. p. 1–8.
- [2] KOHAVI, R. A study of cross-validation and bootstrap for accuracy estimation and model selection. In: **Proc. IJCAI**. [S.l.: s.n.], 1995. v. 14, p. 1137–1143.
- [3] PEDREGOSA, F. et al. Scikit-learn: Machine learning in Python. **Journal of Machine Learning Research**, v. 12, p. 2825–2830, 2011.
- [4] SOKOLOVA, M.; LAPALME, G. A systematic analysis of performance measures for classification tasks. **Information Processing & Management**, v. 45, n. 4, p. 427–437, 2009.
- [5] HUANG, W. et al. Learning urban region representations with POIs and hierarchical graph infomax. **ISPRS J. Photogramm. Remote Sens.**, v. 196, p. 134–145, 2023.